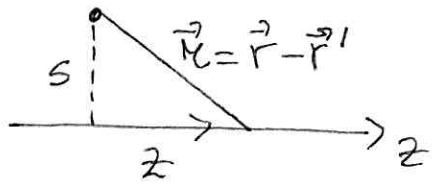
P5.22 $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{\vec{J}(\vec{r}\,')\,dV'}{|\vec{r}-\vec{r}\,'|} = \frac{\mu_0 I \hat{z}}{4\pi}\int \frac{dz}{|\vec{r}-\vec{r}\,'|}$



$$\vec{A} = \frac{\mu_0 I \hat{z}}{4\pi}\int_{z_1}^{z_2} \frac{dz}{\sqrt{s^2+z^2}} = \frac{\mu_0 I \hat{z}}{4\pi} \ln\left(z+\sqrt{s^2+z^2}\right)\Big|_{z_1}^{z_2} =$$

$$= \frac{\mu_0 I \hat{z}}{4\pi} \ln\left(\frac{z_2+\sqrt{z_2^2+s^2}}{z_1+\sqrt{z_1^2+s^2}}\right); \quad \vec{A} = A_z \hat{z}.$$

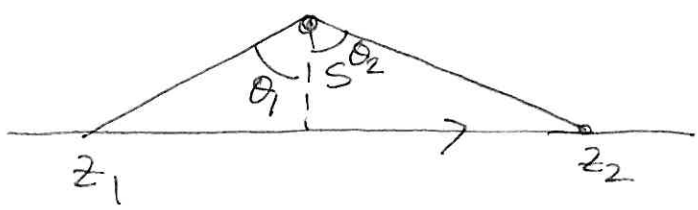
Cylindrical coordinates $\vec{B} = \vec{\nabla}\times\vec{A} = \frac{1}{s}\underbrace{\left(\frac{\partial A_z}{\partial \varphi}\right)}_{=0}\hat{s} - \frac{\partial A_z}{\partial s}\hat{\varphi}$

$$\frac{\partial A_z}{\partial s} = \frac{\mu_0 I}{4\pi}\left(\frac{1}{z_2+\sqrt{z_2^2+s^2}}\,\frac{2s}{2\sqrt{z_2^2+s^2}} - \frac{1}{z_1+\sqrt{z_1^2+s^2}}\,\frac{s}{\sqrt{z_1^2+s^2}}\right)$$

$$= \frac{\mu_0 I s}{4\pi}\left(\frac{z_2-\sqrt{z_2^2+s^2}}{z_2^2-(z_2^2+s^2)}\,\frac{1}{\sqrt{z_2^2+s^2}} - \frac{z_1-\sqrt{z_1^2+s^2}}{z_1^2-(z_1^2+s^2)}\,\frac{1}{\sqrt{z_1^2+s^2}}\right)$$

$$= -\frac{\mu_0 I s}{4\pi s^2}\left(\frac{z_2}{\sqrt{z_2^2+s^2}} - \frac{z_1}{\sqrt{z_1^2+s^2}}\right)$$

$$\Rightarrow \vec{B} = \frac{\mu_0 I}{4\pi s}\left(\sin\theta_2 - \sin\theta_1\right)$$



P5.24 $\vec{B} = \text{const}$. Check $\vec{A} = -\frac{1}{2}(\vec{r} \times \vec{B})$

$$\vec{\nabla} \cdot \vec{A} = \left(-\frac{1}{2}\right) \vec{\nabla} \cdot (\vec{r} \times \vec{B}) = -\frac{1}{2}\left(\vec{B} \cdot \vec{\nabla} \times \vec{r} - \vec{r} \cdot \vec{\nabla} \times \vec{B}\right)$$

(product rule 6)

$$\vec{\nabla} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = 0, \text{ and } \vec{\nabla} \times \vec{B} = 0 \text{ as } \vec{B} = \text{const}$$

$$\Rightarrow \vec{\nabla} \cdot \vec{A} = 0 \quad \underline{\text{ok}}$$

$$\vec{B} = \vec{\nabla} \times \vec{A} = \left(-\frac{1}{2}\right) \vec{\nabla} \times (\vec{r} \times \vec{B}) = -\frac{1}{2}\left((\vec{B} \cdot \vec{\nabla})\vec{r} - (\vec{r} \cdot \vec{\nabla})\vec{B} + \vec{r}(\vec{\nabla} \cdot \vec{B}) - \vec{B}(\vec{\nabla} \cdot \vec{r})\right)$$

(rule 8)

$$= \frac{1}{2}\left\{\left(B_x \frac{\partial}{\partial x} + B_y \frac{\partial}{\partial y} + B_z \frac{\partial}{\partial z}\right)(x\hat{x} + y\hat{y} + z\hat{z}) - \vec{B}\left(\frac{\partial}{\partial x} x + \frac{\partial}{\partial y} y + \frac{\partial}{\partial z} z\right)\right\} =$$

$$= -\frac{1}{2}\left\{\underbrace{B_x \hat{x} + B_y \hat{y} + B_z \hat{z}}_{\vec{B}} - 3\vec{B}\right\} = -\frac{1}{2}(-2\vec{B}) = \vec{B}.$$

It works but is not unique.

P5.25 from P5.22 $\vec{A} = \frac{\mu_0 I \hat{z}}{4\pi} \ln \frac{z_2 + \sqrt{z_2^2+s^2}}{z_1 + \sqrt{z_1^2+s^2}} \Big|_{z_1}^{z_2}$

(a)

for arbitrary z_1 and z_2.

For an infinite wire let $z_1 = -L$, $z_2 = L$ and take the limit $L = \infty$ at the very end.

Then

$$\vec{A} = \frac{\mu_0 I \hat{z}}{4\pi} \ln \frac{\sqrt{L^2+s^2}+L}{\sqrt{L^2+s^2}-L} = \frac{\mu_0 I \hat{z}}{4\pi} \ln \frac{\sqrt{1+\frac{s^2}{L^2}}+1}{\sqrt{1+\frac{s^2}{L^2}}-1} =$$

$$\cong \frac{\mu_0 I \hat{z}}{4\pi} \ln \frac{2+\frac{s^2}{2L^2}}{1+\frac{s^2}{2L^2}} = \frac{\mu_0 I \hat{z}}{4\pi} \ln\left(\frac{4L^2}{s^2}+1\right) = \frac{\mu_0 I \hat{z}}{4\pi} \ln \frac{4L^2}{s^2}.$$

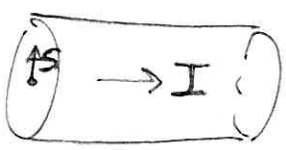
this can safely be neglected for $L \to \infty$.

$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_z}{\partial z} = 0$ because A_z does not depend on z-coordinate.

$$\vec{B} = \vec{\nabla} \times \vec{A} = -\frac{\partial A_z}{\partial s} \hat{\varphi} = \frac{\mu_0 I}{4\pi} \frac{\partial}{\partial s} \ln\left(\frac{s^2}{4L^2}\right) = \frac{\mu_0 I}{4\pi} \frac{2}{s} =$$

$$= \frac{\mu_0 I}{2\pi s}.$$

(b)



Inside the wire

$$B \cdot 2\pi s = \mu_0 I \frac{\pi s^2}{\pi R^2}$$

$$\Rightarrow \vec{B} = \frac{\mu_0 I s}{2\pi R^2} \hat{\varphi}$$

$\vec{B} = B\hat{\varphi} = -\frac{\partial A_z}{\partial s} \hat{\varphi}$, as in this geometry $\vec{A} = A_z \hat{z}$ only.

$$\Rightarrow \frac{\partial A_z}{\partial s} = \frac{-\mu_0 I s}{2\pi R^2} \Rightarrow A_z = -\frac{\mu_0 I}{4\pi R^2}\left(s^2 - d^2\right)$$

here d^2 is some constant.

We can fix constant d by requiring $\vec{A} = A_z \hat{z}$ be continuous at $s = R$:

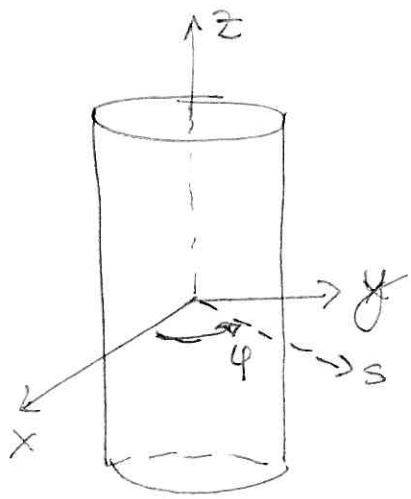
from part (a): $A_z = \frac{\mu_0 I}{2\pi} \ln\left(\frac{2L}{s}\right)\Big|_{s=R} = \frac{\mu_0 I}{2\pi} \ln\left(\frac{2L}{R}\right)$.

from (b): $A_z(s=R) = \frac{\mu_0 I}{4\pi}\left(\frac{d^2}{R^2} - 1\right)$

$$\Rightarrow \frac{\mu_0 I}{4\pi}\left(\frac{d^2}{R^2} - 1\right) = \frac{\mu_0 I}{2\pi} \ln\left(\frac{2L}{R}\right)$$

$$\frac{d^2}{R^2} = 1 + 2\ln\left(\frac{2L}{R}\right).$$

P 6.8



$\vec{M} = k s^2 \hat{\varphi} = M_\varphi \hat{\varphi}$.

Volume magnetization current

$$\vec{J}_b = \vec{\nabla} \times \vec{M} = -\frac{\partial M_\varphi}{\partial z} \hat{s} + \frac{1}{s}\frac{\partial}{\partial s}(s M_\varphi)\hat{z}$$

$$\Rightarrow \vec{J}_b = \frac{1}{s}\frac{\partial}{\partial s}(k s^3)\hat{z} = \frac{3ks^2}{s} = 3ks\,\hat{z}$$

Surface magnetization current $\vec{K} = \vec{M} \times \hat{n} = -ks^2 \hat{z}\Big|_{s=R}$

$\hat{n} = \hat{s}$, $\hat{\varphi} \times \hat{s} = -\hat{z}$

$$\vec{K} = -kR^2 \hat{z}.$$

$\Rightarrow$ surface current flows <u>opposite</u> to <u>bulk</u>.

So the problem is now equivalent to finding $\vec{B}$ due to currents $\vec{J}_b$ and $\vec{K}_b$.

Inside the cylinder $(r<R)$ $\quad \vec{B} = B\hat{\varphi}$

$$B \cdot 2\pi r = \oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc} = \mu_0 \int_0^r (3kr)\, 2\pi r\, dr =$$

$$= \mu_0\, 6\pi k \int_0^r r^2 dr = \mu_0\, 2\pi k\, r^3.$$

$$\Rightarrow B = \frac{2\pi k \mu_0 r^3}{2\pi r} = \mu_0 \underbrace{k r^2}_{M.} \Rightarrow \vec{B} = \mu_0 \vec{M}.$$

<u>$r>R$</u> Outside the cylinder we need total current enclosed:

$$I_{enc} = \int_0^R (3kr)\, 2\pi r\, dr + \underbrace{(-kR^2)(2\pi R)}_{\text{surface current}} =$$

$$= 6\pi k \int_0^R r^2 dr - 2\pi k R^3 = 0$$

$\Rightarrow B = 0$ outside.